

Omkar Shirpure

THIRD YEAR UNDERGRADUATE

Indian Institute of Technology Bombay · Computer Science and Engineering

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Education

Indian Institute of Technology Bombay

Mumbai, India

BACHELOR OF TECHNOLOGY, MAJOR IN COMPUTER SCIENCE AND ENGINEERING

2022 - 2026

Honors & Awards

- 2022 **98.93 percentile**, (AIR 1* OBC-PWD cat.) Joint Entrance Exam **Advanced**, among **150K** candidates
- 2022 **96.87 percentile**, (AIR 3* OBC-PWD cat.) Joint Entrance Exam **Mains**, among **1.5 M** candidates
- 2022 **99.49 percentile**, MHT-CET examination, among **300K** candidates
- 2021 **KVPY Scholarship Awardee**, (AIR 4* PWD cat.) by Indian Institute of Science
- 2020 **NTSE Scholarship Awardee**, by National Council of Educational Research and Training
- 2018 **State Rank 8**, MSCE Scholarship, among **100K** candidates across Maharashtra state

Work Experience

Incoming SWE Intern at Google India Pvt. Ltd.

Summer 2025

LawGPT Team, Flomoney Pvt. Ltd.

Singapore

AI/ML INTERN

May 2024 - Jul. 2024

- Trained and utilized **BART** transformer model with **pipeline** abstraction for title and summary generation, for chronological analysis and insight extraction of the legal documents, improving document retrieval efficiency
- Achieved an accuracy of **95%+** when tested using weighted semantic and syntactic similarity evaluation metrics
- Created a custom Python library for **graph extraction** from documents using **LayoutLMv3** along with **OpenCV**
- Dockerized** and packaged it as a containerized application for future testing and deployment on **Azure Cloud**

Core Development Team, A&PL

USA

SOFTWARE DEVELOPER INTERN, PRO-SPACE

Feb. 2024 - May 2024

- Developed a cross-platform **Flutter** application for both Android and iOS using tech-stack that includes **Node.js** for backend development, **Firebase** for real-time database management, and **Express** for building RESTful APIs
- Integrated **third-party APIs** that include **Firebase** Cloud Messaging, Stripe, Google Maps Services, **OAuth 2.0**
- Stress-tested** systems using **Postman** and **Flutter-test**, evaluating key performance metrics under peak usage

Dream-11 Predictive AI Team Builder

IIT Bombay

IIT BOMBAY CONTINGENT, INTER-IIT TECH MEET 13.0

Nov. 2024 - Dec. 2024

- Preprocessed historical data followed by aggregation and **time-series normalization** for efficient model training
- Developed a **CNN** diffusion model with **spatial-temporal** attention, achieving **20 MAE** in performance prediction
- Integrated **LIME** to provide explainable AI for predictions, offering **feature importance** and decision rationale
- Implemented **Redis-based caching** for LIME explanations, reducing computations and optimizing response time

Projects

Algorithmic Trader

IIT Bombay

GUIDE: PROF. A. GUPTA | COURSE PROJECT: DATA STRUCTURES AND ALGORITHMS

Spring 2024

- Developed a market simulation that uses **threads** to receive orders from several traders and execute them, and trader agents that used strategies such as **median trading** along with **statistical arbitrage** to generate profits
- Designed efficient algorithms for detecting **arbitrage opportunities** in markets by pruning quantities of orders
- Implemented **custom classes** for data structures such as the order trie, red-black trees, and heaps for efficiency

Secure e-Ticket Entry Permit System (STEPS)

IIT Bombay

DEVCOM X ASC HACKATHON

Spring 2024

- Secured **1st** place for creating a foolproof Secure e-Ticket Entry Permit System (STEPS) for events, ensuring authorised entry and exit even **without internet connectivity**, employing **ECDSA encryption** for offline validation
- Implemented **secure environments** within the app to safeguard against tampering, data alteration, and piracy
- Used **HiveDB** local database with **HMAC** encryption ensuring secured **on-device** storage of sensitive user data

IIT Bombay

Spring 2024

- Placed **6th** among **105+ PAN-IIT** teams; most accurate prediction with just **0.4%** deviation from actual solution
- Engineered an advanced **Bayesian Optimization** model using **Gaussian Process Regression** in 3 dimensions
- Incorporated an acquisition function that strategically employs **Expected Improvement** to enhance the predictive accuracy to find minima for an unknown **Black-Box function** considering Exploration-Exploitation trade-off

IIT Bombay

Autumn 2023

- Used **ChampSim** simulator to analyze cache and memory access patterns for various types of Graph Algorithms
- Generated and plotted data by running traces on different architecture combinations with **30 M+** instructions
- Improved **IPC** in graph workloads by developing a combination of exclusive and non-inclusive cache hierarchies

IIT Bombay

Spring 2023

- Developed a **multi-threaded** web-scraper, supporting **recursive** scraping across multiple URLs simultaneously
- Designed a **master-worker** architecture using multi-threading, with inter-thread communication via **Shared Memory IPC**, facilitating efficient task distribution, and synchronization for scalable, parallelized data scraping

IIT Bombay

Summer 2024

- Implemented **NEAT** algorithm and developed **Pygame** simulations for autonomous flight training of the drones
- Configured **protocols** for decentralized decision-making in swarm operations to complete the tasks efficiently
- Performed comparative analysis of models utilizing **Kalman estimation** and **genetic algorithm-based training**.

Extracurricular Activity

Jan. 2023 - Mar. 2024

- Part of a **25+** member team responsible for designing, developing, and maintaining multiple institute projects
- **InstiApp** : Over **10,000+** downloads and **3,000+ MAUs** meeting info-social necessities of IIT Bombay campus
 - Developed **Buy-n-Sell** feature, facilitating seamless P2P transactions of secondhand goods among students
- **Mess-I** : Digitalized mess management platform serving over **10,000+** students daily across 10+ campus hostels
 - Implemented a **remote configuration** feature facilitating remote debugging and troubleshooting anomalies

2023, 2024

- **2024:** Placed in the **Top 3** among **250+ PAN-IIT** teams for developing an efficient, and accurate **QnA model** using **s-BERT**, also providing an answer summary using **BART**, while ensuring scalability with **Docker** and **Kubernetes**
- **2023:** Placed in the **Top 10** among **120+ PAN-IIT** teams for proposing design for an efficient healthcare service

Skills

Programming	C/C++, Python, Go, Scala, Haskell, Lisp, Bash, x86 and MIPS Assembly, VHDL, DART, Verilog
Development	React, Next.js, Flutter, Node.js, Django, MonogoDB, PostgreSQL, Angular
Data Science	Excel, NumPy, Pandas, MatPlotLib, SciPy, PyTorch, TensorFlow, scikit-learn
Utilities	Kubernetes, Docker, Redis, ELK Stack, HiveDB, , MATLAB, Git, Sed, Awk, GDB

Relevant Coursework

- * Data Structures and Algos
- * Data Analysis & Interpretation
- * Computer Architecture
- * Computer Networks
- † Compilers
- * Analysis of Algorithms
- * Software Systems Lab
- * Operating Systems
- * Abstractions and Paradigms
- † Cloud Computing
- * Discrete Structures
- * AI and ML
- * Logic and TOC
- * Image Processing
- † Database Systems

*: Completed

t: To be completed by Spring 2025

Miscellaneous

- Assisted in **logs analysis** at **Computer Center, IIT Bombay**, leveraging the **ELK Stack** to scrutinize log alerts
- Curated intuitive, activity-based questionnaires impacting **200+** students under the **National Service Scheme**
- Secured **A grade** in both Elementary and Intermediate Maharashtra State Drawing Examinations in 2018, 2019
- **Organizer** in Marketing Division of E-Summit 2023, the Flagship event of **Entrepreneurship Cell, IIT Bombay**